

AI Stock Monitoring Multi-Agent System

System Design Document

1. Project Overview

Objective

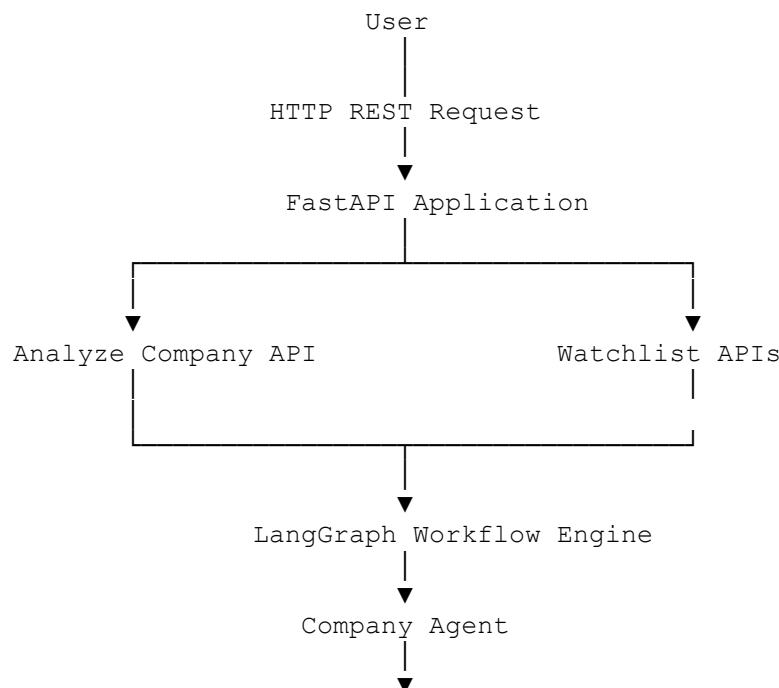
The AI Stock Monitoring Multi-Agent System automatically analyzes publicly traded companies and generates AI-powered investment reports.

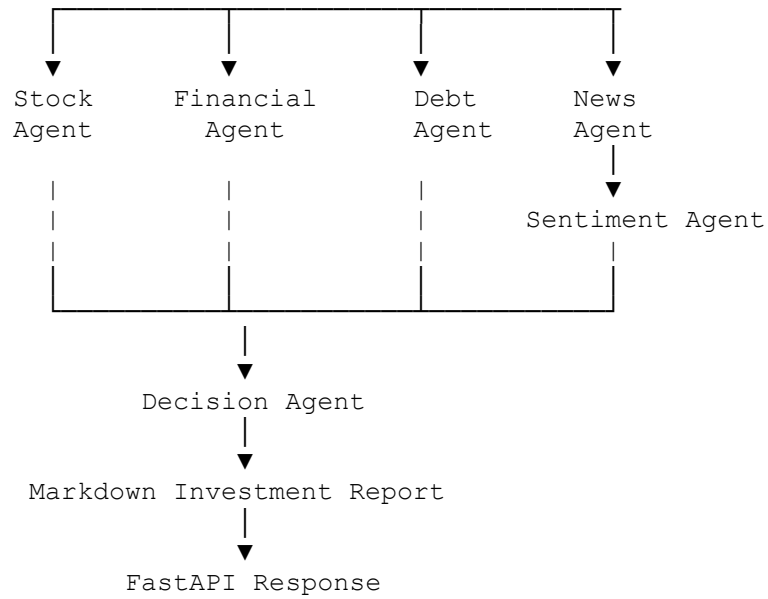
The system integrates:

- FastAPI
- LangGraph
- Groq Llama 3.3 70B
- Yahoo Finance
- Google News RSS
- APScheduler

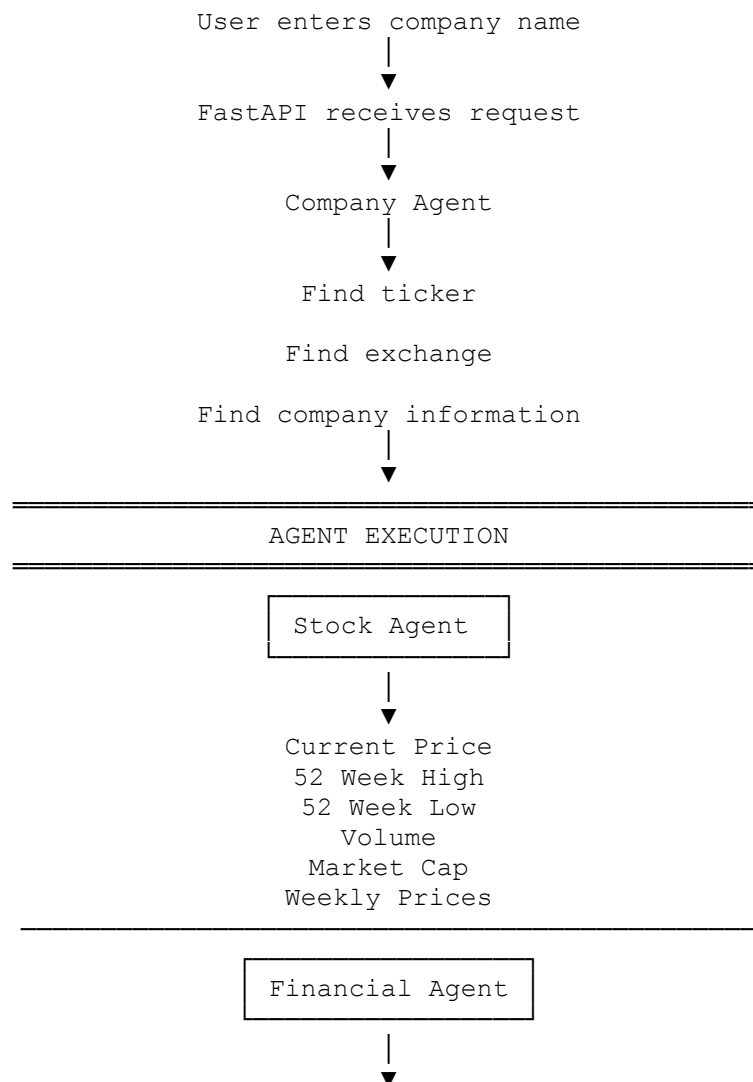
The system monitors companies in a watchlist and periodically generates professional stock analysis reports.

2. Overall System Architecture





3. Complete Workflow



PE Ratio
EPS
Revenue Growth

Debt Agent

Debt to Equity
Current Ratio
Liquidity

News Agent

Latest Headlines
Google News RSS

Sentiment Agent

Analyzes all retrieved news

Returns

Positive

Neutral

Negative

Decision Agent

Receives

✓ Company Information

✓ Stock Information

✓ Financial Information

✓ Debt Information

✓ News Headlines

✓ Sentiment Analysis

Groq Llama 3.3

Professional Markdown Report

BUY/ HOLD/ SELL WITH Confidence Score

Detailed Reasoning

4. Component Architecture

stock_ai_agent/



5. Agent Responsibilities

Company Agent

Purpose

- Search company
- Identify ticker
- Get Exchange
- Get Company information

Example:

Input:

Apple

Output:

AAPL

Apple Inc.

NASDAQ

Stock Agent

Returns following by collecting data from yahoo finance

- Current Price
 - Previous Close
 - Day High
 - Day Low
 - 52 Week High
 - 52 Week Low
 - Market Capitalization
 - Trading Volume
 - Average Volume
 - Dividend Yield
 - Beta
 - Weekly Closing Prices
-

Financial Agent

Returns following by collecting data from yahoo finance

- PE Ratio
 - EPS
 - Revenue Growth
-

Debt Agent

Returns following by collecting data from yahoo finance

- Debt to Equity
 - Current Ratio
 - Liquidity
-

News Agent

Collects: Latest news using Google News RSS

Returns: Top 5 Headlines About Company

Sentiment Agent

Uses: Groq Llama 3.3

Input: News Headlines

Output: Positive/ Neutral/ Negative

Decision Agent

Receives outputs from every previous agent.

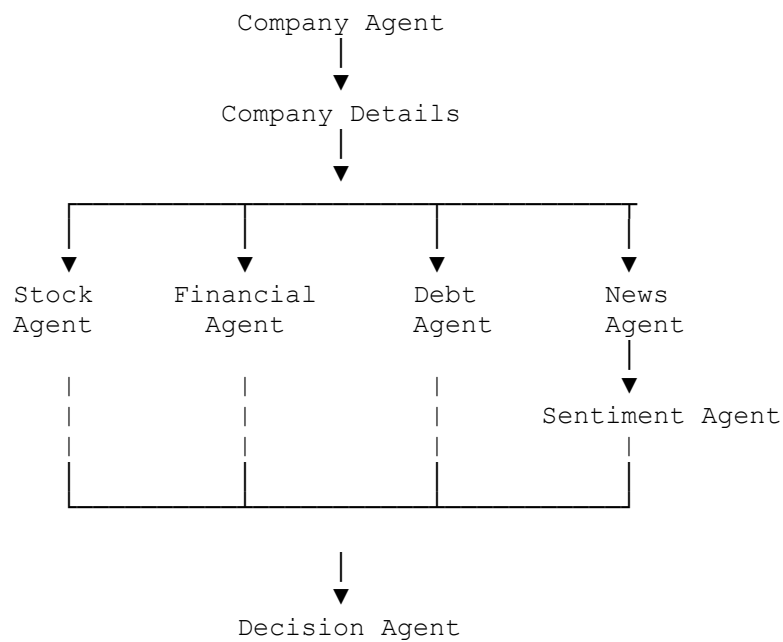
Creates report in markdown format which includes

- Company Overview
 - Market Position
 - Financial Analysis
 - Debt Analysis
 - News Analysis
 - Opportunities
 - Risks
 - BUY/HOLD/SELL Recommendation
 - Confidence Score
 - Detailed Explanation
-

6. Parallel Agent Design

The Company Agent is the only mandatory first step because every other agent requires the ticker symbol.

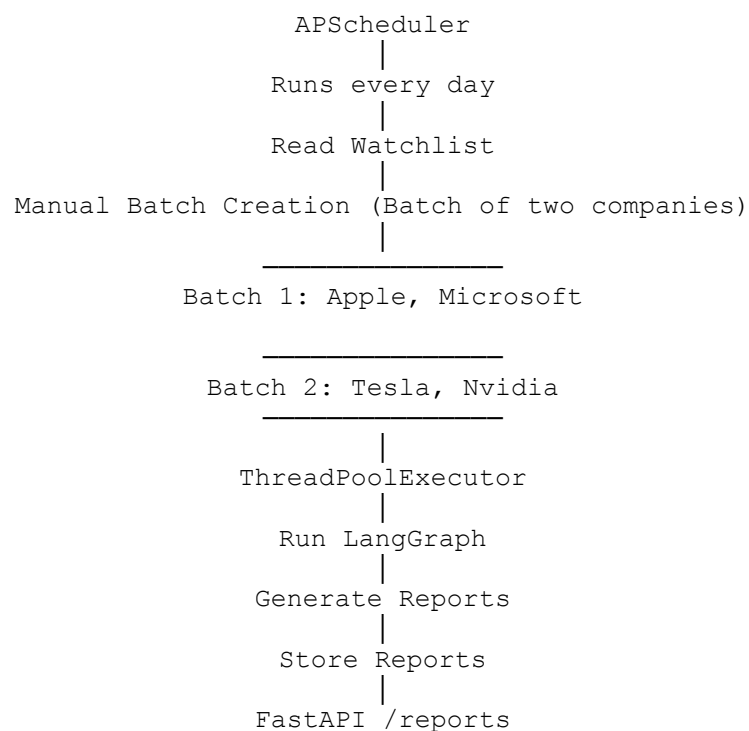
After retrieving the ticker, the remaining retrieval agents work independently.



These agents do not depend on each other except sentiment agent (sentiment agent is dependent on results of news agent, making them ideal candidates for parallel execution.

7. Scheduler Design

The scheduler automatically analyzes companies in the watchlist.



8. Manual Batch Handling

Instead of

```
Apple
↓
Microsoft
↓
Tesla
↓
Nvidia
```

The scheduler creates batches.

```
Batch Size = 2
↓
Apple
Microsoft
↓
Tesla
Nvidia
```

Each batch executes using:

```
ThreadPoolExecutor
```

Benefits:

- Faster execution
- Better scalability
- Lower API latency
- Controlled concurrency

9. API Endpoints

Add Company

```
POST /watchlist/{company}
```

Remove Company

```
DELETE /watchlist/{company}
```

View Reports

```
GET /reports
```

11. Technology Stack

Layer	Technology
Backend	FastAPI
Workflow	LangGraph
LLM	Groq Llama 3.3 70B
Financial Data	Yahoo Finance
News	Google News RSS
Scheduler	APScheduler
Parallel Processing	ThreadPoolExecutor

12. Future Enhancements

- Real parallel execution using LangGraph fan-out/fan-in.
 - Technical Analysis Agent (RSI, MACD, Bollinger Bands, Moving Averages).
 - Portfolio Management Agent.
 - Email, Whatsapp and Telegram alert notifications.
 - WebSocket-based real-time dashboard updates.
 - PostgreSQL for production deployments.
 - Historical report storage with vector databases for RAG-based querying.
 - Docker and Kubernetes deployment.
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13. Conclusion

The AI Stock Monitoring Multi-Agent System follows a modular, extensible architecture where each agent specializes in a single responsibility. The Company Agent first resolves the company identity, after which the Stock, Financial, Debt, and News agents can execute in parallel to gather independent information. The Sentiment Agent analyzes the collected news, and finally the Decision Agent aggregates all outputs to produce a comprehensive Markdown investment report with a BUY, HOLD, or SELL recommendation.

This design supports scalable data retrieval, periodic monitoring through APScheduler, manual batch processing with `ThreadPoolExecutor`, and clean separation of responsibilities, making it well suited for future enhancements such as real-time alerts, technical analysis, portfolio management, and production deployment.